Sample Data / Test cases:

Integrated Solar Utility System for Sustainable Rural Communities (ISUS)

1. Introduction

This document presents the sample simulation parameters and test procedures used to evaluate the electrical, thermal, and water-production performance of the Integrated Solar Utility System for Sustainable Rural Communities.

2. Sample Simulation Data

The TEG simulation uses four modules, a Seebeck coefficient of 0.025 V/K, a hot-side temperature of 692 K, and a cold-side temperature of 238 K. A 10 Ω resistive load is used in the reported TEG configuration, which produces an output voltage of 30 V.

The boost converter uses a 30 V input, a 1 mH inductor, a 470 μF capacitor, and a 280 Ω load resistance. The reported output voltage is 37 V.

The buck converter uses a 30 V input, a 1 mH inductor, a 470 μF capacitor, and a 32 Ω load resistance. The reported output voltage is 6 V.

These values are supplied simulation results and should be validated against the original simulation files before being treated as reproducible results.

3. Test Cases

Test Case 1: Solar Panel Output

Objective: To evaluate the electrical output of the 20 W, 12 V solar panel.

Procedure: Expose the panel to sunlight and measure the output voltage and current under recorded environmental conditions.

Expected Result: The output should be consistent with the panel specifications and actual solar conditions.

Test Case 2: TEG Output

Objective: To evaluate the TEG electrical output.

Procedure: Run the TEG simulation using the specified model parameters and observe the output waveform.

Expected Result: The simulation should reproduce an output close to the reported 30 V under the stated conditions.

Test Case 3: Boost Converter

Objective: To evaluate the voltage-boosting operation.

Procedure: Apply the specified 30 V input and observe the simulated output voltage.

Expected Result: The output should approach the reported simulation value of 37 V.

Test Case 4: Buck Converter

Objective: To evaluate the voltage-reduction operation.

Procedure: Apply the specified 30 V input and observe the simulated output voltage.

Expected Result: The output should approach the reported simulation value of 6 V.